

Total No. of Questions : 8]

SEAT No. :

PD-4855

[Total No. of Pages : 2

[6404]-386

B.E. (Artificial Intelligence & Data Science)

DISTRIBUTED COMPUTING

(2019 Pattern) (Semester - VIII) (417531)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.

- Q1)** a) Explain the key differences between Centralized and Distributed Load Balancing Techniques. **[6]**
- b) Describe what Consensus algorithms are and explain how one specific algorithm works. **[6]**
- c) Compare and contrast how Weighted Round Robin and Least Connection load balancing algorithms would perform in a system with varying server capacities and network loads. **[6]**

OR

- Q2)** a) Describe one variant of Paxos and explain how it differs from the original Paxos algorithm. **[6]**
- b) Describe how Genetic Algorithms can be used for task scheduling and explain their benefits in this context. **[6]**
- c) Apply load balancing and resource allocation strategies to optimize performance in a cloud computing environment. How would you implement these strategies to handle varying workloads? **[6]**

P.T.O.

- Q3)** a) Explain Elastic Averaging SGD. [9]
b) Explain Systems and Architectures for Distributed Machine Learning. [8]

OR

- Q4)** a) What is Apache Spark? Explain the working of Apache Spark. [9]
b) Explain i) Federated Learning.
ii) Elastic Averaging SGD. [8]

- Q5)** a) How would you apply Secure Multi-Party Computation (SMPC) to protect sensitive data in a collaborative machine learning task? [6]
b) What are the key differences between SIMD and MIMD? [6]
c) Describe the concepts of Threat Hunting and Visualization and explain how they are used in cybersecurity. [6]

OR

- Q6)** a) Summarise the AI-based Intrusion Detection and Threat Mitigation Techniques. [6]
b) Explain Anomaly as well as Behavior AI-based Intrusion Detection & Threat Mitigation Techniques [6]
c) Discuss about various types of real-time analytics used in distributed computing systems. [6]

- Q7)** a) Explain how reinforcement learning can be applied to load balancing in distributed systems, highlighting its potential advantages. [9]
b) Apply different Big Data processing frameworks in a distributed computing environment to solve a specific data analysis problem. How would you choose and implement the appropriate framework? [8]

OR

- Q8)** a) Describe the different types of data ingestion and explain how each type is used in data processing workflows. [9]
b) Describe AI-based intrusion detection and threat mitigation techniques and explain how they help enhance network security. [8]

